

Custom Marker Specification & Generation Logic

To ensure highly robust and instantaneous detection, the computer vision engine is programmed to expect a strict set of geometric rules for any valid custom marker.

1. Generation Logic & Measurements

All custom markers (whether Type 1 or Type 2) must adhere to the following geometric ratios. These are defined mathematically based on the total width/height of the printed marker square ($W \times H$):

- Outer Shape:** The marker must be a square ($W \times H$).
- Solid Border Thickness:** The black border around the marker must occupy exactly **5% of the total width** ($0.05 \times W$).
- Anchor Dot Location (Type 1):** To indicate orientation, a square "anchor dot" must be placed in exactly one corner.
- Anchor Dot Inset:** The anchor dot must start exactly **18% inward** from the absolute corner of the marker ($0.18 \times W$).
- Anchor Dot Size:** The anchor dot must be a solid black square with a width/height equal to **15% of the total marker width** ($0.15 \times W$).
- Negative Space Requirement:** To prevent false positives, at least two of the other three corners must be completely empty (less than 8% ink coverage).

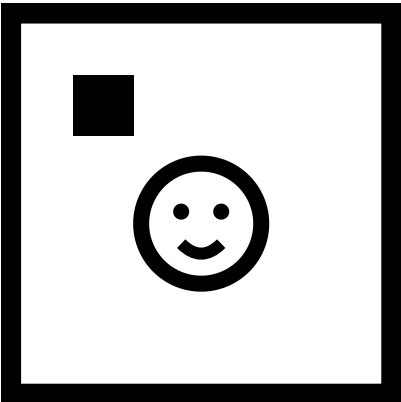
2. Test Images in Different Orientations

Below is the digitally generated master template for the Type 1 Custom Marker, rendered in all four 90-degree rotational orientations.

*No matter which of these orientations the camera captures, the algorithm mathematically calculates the `targetQuad` and forcefully rotates the image matrix to bring the anchor dot perfectly into the **Top-Left** (0° rotation) canonical position.*

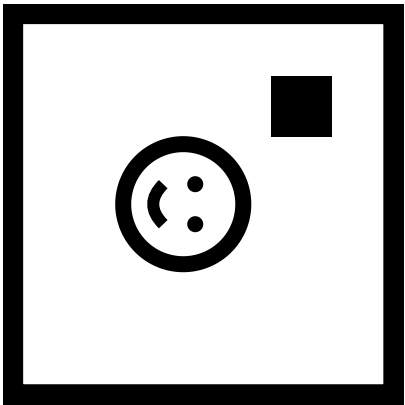
Canonical Upright (0° Rotation)

The canonical upright orientation places the anchor dot exactly in the **Top-Left** quadrant.



90° Clockwise Rotation

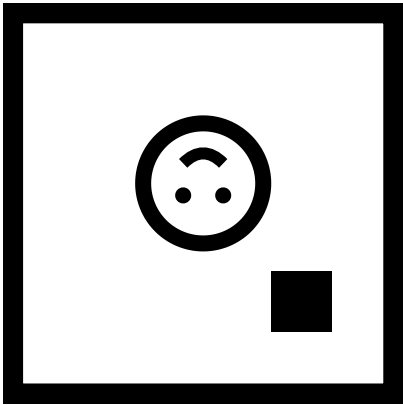
The anchor dot shifts to the **Top-Right**. The engine detects this and rotates the matrix 90° counter-



clockwise.

180° Rotation (Upside Down)

The anchor dot shifts to the **Bottom-Right**. The engine detects this and rotates the matrix 180°.



270° Clockwise Rotation

The anchor dot shifts to the **Bottom-Left**. The engine detects this and rotates the matrix 90° clockwise.

